Lists in Python

Python Lists are just like dynamically sized arrays, declared in other languages (vector in C++ and ArrayList in Java). In simple language, a list is a collection of things, enclosed in [ ] and separated by commas.

- ⇨ Mutable: Change their content without changing their identity.

Lists are the simplest containers that are an integral part of the Python language. Lists need not be homogeneous always which makes it the most powerful tool in Python. A single list may contain DataTypes like Integers, Strings, as well as Objects. Lists are mutable, and hence, they can be altered even after their creation.

Syntax for creating a list:

Listname = [ele1 , ele2 , ele3 , … .. , ele_n]

(or)

Listname = list((ele1,ele2,….))

Methods in list

Function	Description
Append()	Add an element to the end of the list
Extend()	Add all elements of a list to another list
Insert()	Insert an item at the defined index
Remove()	Removes an item from the list
Clear()	Removes all items from the list
Index()	Returns the index of the first matched item
Count()	Returns the count of the number of items passed as an argument
Sort()	Sort items in a list in ascending order
Reverse()	Reverse the order of items in the list
copy()	Returns a copy of the list

Example:

```
>>> l=[]
>>> type(l)
<class 'list'>
>>> x=[1,2,"hello",9.54]
>>> x[3]
9.54
>>> x[-2]
'hello'
>>> x[-1]
9.54
>>> x[ : ]
[1, 2, 'hello', 9.54]
>>> x[2:4]
['hello', 9.54]
>>> x
[1, 2, 'hello', 9.54]
>>> x.append(20)
>>> x
[1, 2, 'hello', 9.54, 20]
>>> y=[10,20,30]
>>> y
[10, 20, 30]
>>> x.extend(y)
```

```
>>> x
[1, 2, 'hello', 9.54, 20, 10, 20, 30]
>>> x.insert(5,"Hi")
>>> x
[1, 2, 'hello', 9.54, 20, 'Hi', 10, 20, 30]
>>> x.index(10)
6
>>> y
[10, 20, 30]
>>> y.clear()
>>> y
[]
>>> y=x.copy()
>>> y
[1, 2, 'hello', 9.54, 20, 'Hi', 10, 20, 30]
>>> y.count(20)
2
>>> len(x)
9
>>> y.pop()
30
>>> y.remove(20)
>>> y
[1, 2, 'hello', 9.54, 'Hi', 10, 20]
>>> y.reverse()
```

```
>>> y
[20, 10, 'Hi', 9.54, 'hello', 2, 1]
>>> x=[1,8,3,4,0,12,22,90,2,3,4,4]
>>> x.sort()
>>> x
[0, 1, 2, 3, 3, 4, 4, 4, 8, 12, 22, 90]
>>> y=["a","m","b","k","c"]
>>> y.sort()
>>> y
['a', 'b', 'c', 'k', 'm']
>>> del y
>>> y
Traceback (most recent call last):
  File "<pyshell#38>", line 1, in <module>
    y
NameError: name 'y' is not defined
```

Tuples in python

They are sequence just like a list.

Ordered and immutable.

Written inside round drackets ➔ () and separated with commas.

No modifications can be done in tuple.

Syntax:

T_n=(e1,e2,….,en)

(or)

T_n = tuple((e1,e2,….))

- ⇨ tn = ele1, ele2, …..#Tuple packing.
- ⇨ Ex:

  X=(1,2,3)

  (a,b,c)=X

  #Tuple unpacking.

Single element inside a tuple can be created as ➔ tn=(ele,)

Example:

```
>>> D=(1,2,3,"h")

>>> l=list(D)

>>> l

[1, 2, 3, 'h']

>>> l.append(47)

>>> l
```

```
[1, 2, 3, 'h', 47]
>>> D=tuple(l)
>>> D
(1, 2, 3, 'h', 47)
>>>
>>> D=(1,2,3,4)
>>> D
(1, 2, 3, 4)
>>> D.index(3)
2
>>> D=(1,2,2,2,3,4,5,3,2,2)
>>> D.count(2)
5
>>> min(D)
1
>>> max(D)
5
>>> len(D)
10
>>> D.clear()
Traceback (most recent call last):
  File "<pyshell#74>", line 1, in <module>
    D.clear()
AttributeError: 'tuple' object has no attribute 'clear'
>>> del D
```

```
>>> D
Traceback (most recent call last):
  File "<pyshell#77>", line 1, in <module>
    D
NameError: name 'D' is not defined
```